

Every layer of evidence, one ranked list of targets, and the paper trail to defend it

TerraPulse takes an exploration project's data — yours and the world's public record — and returns **ranked drill targets, each carrying the evidence that produced it** and the full lineage needed to stand behind it. 32 features across seven groups, twelve integrated data sources, built for teams who have to justify where the rig goes.

The problem, as exploration teams actually experience it

The data doesn't meet

Magnetics in one format, soil geochem in another, assay certificates as PDFs, imagery and geology somewhere else entirely. Getting them onto one grid is senior GIS work measured in weeks.

The reasoning doesn't survive

A target is a judgement made by an experienced geologist looking at several screens. When they leave, or the board asks why, the reasoning is gone — only the conclusion remains.

The work doesn't re-run

New assays land and the integration starts again. The last analysis was a one-off deliverable, not something you can point at fresh data and press a button on.

Three claims the platform makes

Everything else is in service of these.

01 Assembly is automatic

Upload what you hold; the platform fetches the rest. Sentinel-2 imagery, Copernicus terrain, NOAA regional magnetics, Macrostrat and OneGeology lithology, USGS known deposits, protected areas and road networks are pulled to your licence boundary and put on the same grid as your own survey data. What took weeks of GIS assembly is a boundary polygon and a wait.

02 The ranking explains itself

Every target carries per-target evidence attribution: which layers pushed the score up, and by how much. Not a heat map you interpret — a ranked list that tells you what it was reasoning from, with a depth-to-source estimate attached. A geologist can agree or disagree with a specific number rather than with a colour.

03 The output is defensible

Every layer records its source file, checksum, processing steps and algorithm parameters. Every report cites what it drew on and is versioned rather than overwritten. Qualified-Person notes and review status attach to targets and follow them into exports. Built for the moment someone asks you to show your working.

How it works, end to end

One pipeline, five stages. Each stage writes provenance, so the last stage can always answer for the first.

STAGE 1

Ingest

Your survey data, validated and checksummed.

- Geophysics: XYZ, GRD, DAT, TIF
- Geochemistry: CSV
- Assay certificates: PDF

STAGE 2

Augment

Public sources fetched to your boundary.

- Sentinel-2 alteration indices
- Terrain, regional magnetics
- Geology, deposits, roads, reserves

STAGE 3

Normalise

Everything onto one comparable grid.

- Kriged interpolation to 50 m
- Per-element percentile classes
- Reduction to pole, depth estimation

STAGE 4

Score & rank

Weighted deposit model over the evidence grid.

- Score per cell, 0–100
- Clustered to distinct targets
- Evidence contribution per target

STAGE 5

Explain & export

Something you can act on and defend.

- Reports by audience persona
- PDF, CSV, KML, shapefile
- Full lineage per layer and target

The full feature set

Seven groups covering the MVP, plus five platform systems added since. Phrased by what each one gets you rather than what it is.

Projects, teams and isolation GROUP 1 · 5 FEATURES

Multi-tenant from the database up, so a company and a sole consultant run on the same product without either seeing the other.

Account & sign-in	Email and password, short-lived access tokens with refresh. Permissions travel in the token as a module list.
Project setup	Name, commodity, country, and a licence boundary either drawn on the map or uploaded as GeoJSON. The boundary scopes everything downstream.
Project dashboard	Status, target count, area and last activity across every project you hold.
Team invites	Invite by email into Editor or Viewer bundles, or a custom module set. Seats enforced against the plan at invite time.
Tenant isolation	Row-level security in the database, tenant-prefixed object storage, module checks on every route. A bug in a query still cannot cross tenants.

Data in, processed properly GROUP 2 · 6 FEATURES

The unglamorous half of the product, and the half that decides whether the rest is trustworthy.

Geophysics upload	Drag and drop up to 500 MB. Content is validated, not just the extension, and every file is checksummed on arrival.
Processing pipeline	Parse, reproject, krigé to a regular grid, reduce to pole, estimate depth to source, write a cloud-optimised raster and register it as a map layer — recording parameters at each step.
Satellite auto-fetch	Pulls the most recent low-cloud Sentinel-2 scene over your boundary and derives iron-oxide, clay/hydroxyl and vegetation indices. The alteration ratios feed scoring.
Geochemistry normalisation	Column auto-detection, unit handling, deduplication, then 85th/95th/99th percentile thresholds computed per element from your own population rather than fixed cut-offs.
Live job progress	Stage-by-stage progress while work runs, with plain-English failures instead of stack traces.
Assay PDF extraction	Reads sample IDs and assay tables out of lab certificates and links them to spatial sample points by ID match.

The map GROUP 3 · 4 FEATURES

Where the evidence is actually read. Everything is a layer you can turn on, fade, interrogate and trace back to its source.

Multi-layer viewer	Five basemaps including satellite and 3D terrain, your licence boundary always drawn, every processed layer available as an overlay.
Layer panel	Visibility and opacity per layer, one-click hide-all, and a hover explanation of what each layer is, why it matters and how to read it.
Add layer on demand NEW	Fetch a public dataset — imagery, terrain or regional magnetics — straight from the map, without leaving for the upload screen.
Click to inspect	Any point returns its coordinates in both WGS84 and local UTM, raster values for active layers, and the nearest sample and known deposit.
Anomaly display	Geochemistry coloured by percentile class, from background through weak, moderate and strong, with every element value one click away.

Analysis and ranked targets GROUP 4 · 7 FEATURES

The core. An expert-weighted deposit model over an assembled evidence grid, and an answer that shows its reasoning.

Run analysis	One action over whatever evidence the project holds, with progress reported while it runs.
Evidence grid	A regular grid over the licence where each cell carries its magnetic value and gradient, alteration indices, geochemical anomaly scores, lithology, slope, road distance, deposit proximity and protected-area status.

Prospectivity scoring	An orogenic-gold model with published weights — magnetic low, structural intersection, iron oxide, clay alteration, gold geochemistry, deposit proximity — producing a 0–100 score per cell.
Prospectivity surface	The scored grid as a map layer, so the ranking can be read in its spatial context rather than as a table alone.
Ranked drill targets	Highest-scoring cells clustered so you get distinct targets rather than duplicates of one anomaly, each with rank, score, coordinates in both systems and a depth-to-source estimate.
Provenance panel	"How was this calculated?" per layer and per target: datasets used with checksums, every processing step, the evidence contributions, and the model version.
Qualified-Person annotation	Notes, review status, reviewer name and date on each target — persisted, and carried into every export.

Exploration Copilot

GROUP 5 · V2 · 14 TOOLS

An assistant that can only read your project, through named tools, and has to cite what it used. Designed so that it cannot invent a number.

Grounded question answering	Ask about targets, geochemistry, layers, coverage or the area of interest and get an answer assembled from actual project data, with citations.
Interpretation on request	Why the top targets ranked where they did, what the geochemistry is saying, what the structural setting looks like, and where the data has gaps.
Charts from live data	Renders figures — target scores, evidence drivers, anomaly classes, percentile matrices — where every value is resolved server-side from the database, never written by the model.
Map control from chat	Ask to show or hide a layer and the map view changes.
Reads your reports NEW	Lists what has been generated and reads a report's full text, so questions about past conclusions are answered from the document rather than from memory.
Verified claims	Deposit names in any generated summary are checked against a known-deposits register before the text is shown. Unverified names are stripped.
Sessions and history	Named conversations per project, with earlier exchanges available as context.

Reporting and export

GROUP 6 · V2 · 5 TYPES × 5 PERSONAS

The deliverable. Composed from the same data the map and the targets use, so the document and the screen can never disagree.

Five report types	Technical Exploration, Executive Summary, Drill Program, Data QA and Comprehensive — each a curated set of sections, with sections that don't apply to a project skipped rather than padded.
Five audience personas	The same facts written for an exploration geologist, a Qualified Person under NI 43-101/JORC conventions, an investor, a project manager or an environmental officer. QP and investor voices additionally strip unsupported claims.
Versioned, not overwritten	Regenerating produces a new version and preserves the old one. Sections move draft → review → final and record whether a human edited them.
Export formats	PDF for circulation, shapefile and KML for GIS and handhelds, CSV for coordinates and scores.
White-label	Consultants can put their own logo, company name and contact details on every generated document.

Field use and portfolio view

GROUP 7 · 4 FEATURES

Exploration happens where connectivity doesn't. The product is built to degrade rather than stop.

Offline map tiles	The current view and surrounding zoom levels are cached on project load, so the map still draws without a connection.
Offline targets and notes	Targets and their evidence stay readable offline, and a Qualified-Person note written in the field is queued on the device and synced on reconnect.
Connectivity state	A persistent indicator, with the actions that genuinely need a network disabled and explained rather than failing silently.
Portfolio dashboard	Every project at once: total ranked targets, total licence area, and breakdowns by commodity and status.

Platform systems

V2 · OPERATOR-FACING

What makes it a product rather than a tool: it can be configured, metered and operated without a developer.

Pluggable AI providers	Anthropic, OpenAI or Gemini, configured per feature — nine separate jobs, each pointed at whichever model suits it. Keys are stored encrypted. Anything unconfigured falls back to deterministic output rather than failing.
Plans and entitlements	Four plans gating modules, seats, projects, analysis runs, report generations, storage and rate limits.
Usage-based credits	A hybrid model: fixed cost for an analysis run, token-based for AI work, with top-ups through Paystack and a visible ledger.
Admin portal	Tenants, plans, credit costs, AI providers, report personas, basemap catalogue, usage, application config with an audit trail, live logs and error tracking.
Notifications and activity	A per-tenant feed of what ran, what finished and what failed, per project and across the organisation.
Guided onboarding NEW	A new account gets a fully worked example project — real data through the real pipeline, with ranked targets and a drafted report — plus an optional guided tour of building their own.

Why this is defensible, not just automated

The argument that matters to a technical buyer. Automation is easy to claim; the constraints below are what make the output usable in a disclosure context.

Four deliberate constraints

- **The model cannot write a number.** Every figure and statistic in the Copilot and in reports is resolved from the database server-side. The language model selects and interprets; it never supplies values.
- **Ranking is withheld where the model does not apply.** The deposit model is tuned for orogenic gold. Other commodities get full data, maps and layers — but no ranked targets, rather than numbers from the wrong model.
- **Deposit names are verified before they are shown.** Any analogue cited in a generated summary is checked against a known-deposits register, and stripped if it does not match.
- **Nothing is a black box.** Weights are published, contributions are per-target, processing parameters are recorded, and source files are checksummed.

Where the data comes from

Twelve integrated sources, all licensed for commercial use, all fetched to your boundary.

SOURCE	PROVIDES	ROLE IN SCORING
Your survey data	Magnetics, geochemistry, assays	Primary anomaly evidence
Sentinel-2	Iron oxide, clay/hydroxyl, vegetation	Alteration signature
Copernicus DEM	Slope, hillshade, drainage	Terrain and access
NOAA EMAG2	Regional magnetic anomaly	Basement and structural context
Macrostrat	Lithology, age, structural units	Host-rock favourability
OneGeology / BRGM	Geological map context	Interpretation backdrop
USGS MRDS	Known deposits and occurrences	Proximity weighting, claim verification
Protected areas (WDPA)	Conservation boundaries	Exclusion constraint
OpenStreetMap	Road network, settlements	Access and logistics scoring
Landsat / ASTER	Long-archive SWIR	Change detection, alteration mapping

Commercial shape

Plans gate modules and volume; AI work additionally draws on credits, so heavy users pay for what they use rather than being throttled.

PLAN	SEATS	PROJECTS	ANALYSIS RUNS	REPORTS	INCLUDES
Starter	2	3	20	10	Map, upload, analysis, reports
Pilot	10	15	150	60	+ Copilot, team

Pro	8	20	100	50	+ Copilot, team
Enterprise	100	200	1,000	500	+ Copilot, team

Monthly limits, all editable per tenant. Pilot exists precisely for an engagement like this one — near-Pro capability, sized for a real evaluation rather than a trial.

What it doesn't do yet

Say these before they're found. Named by you they read as engineering judgement; discovered by them they read as overselling.

Ranking is gold-only

Copper, lithium, nickel and the rest get every layer, map and geochemical treatment, but no ranked targets. The weights and lithology favourability are tuned for orogenic gold, and returning numbers from the wrong deposit model would be worse than returning none. Per-commodity profiles are specified and not yet built — offer a date.

A boundary is required

Imagery, terrain, regional magnetics and analysis all need a licence polygon. Without one they refuse rather than guess at an extent.

Smaller gaps

Social sign-in, self-service password change, and branding accent colour and font are visible in the interface but not yet functional.

Feature set verified against the codebase and the MVP specification · pair with the demo run sheet for the live segment